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Lab: Concurrency Control

Lab 1: Experiencing Blocking

The screenshot displays two side-by-side PostgreSQL query editors, both connected to a database named 'first_data' on a 'postgres@PostgreSQL 18' instance. The left editor shows a transaction T1 with the following SQL statements:

```
1 --T1
2 --Step1
3 begin;
4 select film_id, rental_rate
5 from film where film_id=1;
6
7 update film set rental_rate=4.99
8 where film_id=1;
9
10 --Step3
11 commit;
12 --Release the lock
```

The right editor shows a transaction T2 with the following SQL statements:

```
1 --T2
2 --Step2
3 begin;
4 select film_id, rental_rate
5 from film where film_id=1;
6 update film set rental_rate=5.99
7 where film_id=1;
8
9 --Step4
10 commit;
```

Below the query editors, the 'Data Output' and 'Messages' tabs are visible. The left editor's output shows 'UPDATE 1' and 'Query returned successfully in 23 secs 700 msec.' The right editor's output shows 'COMMIT' and 'Query returned successfully in 73 msec.' Two green notification boxes are present: one for the left query stating '✓ Query returned successfully in 23 secs 700 msec. ✕' and another for the right query stating '✓ Query returned successfully in 73 msec. ✕'. The status bar at the bottom indicates the total rows and query completion time for each editor.

Lab 3: Creating a Deadlock

The screenshot displays two side-by-side PostgreSQL query editors. The left editor shows a sequence of SQL statements: Step 1 (BEGIN), Step 2 (UPDATE film SET rental_rate = 2.99 WHERE film_id = 100), Step 3 (UPDATE film SET rental_rate = 3.99 WHERE film_id = 200), and Step 4 (UPDATE film SET rental_rate = 3.99 WHERE film_id = 200). The right editor shows the same sequence, but with a deadlock error message: "ERROR: deadlock detected. Process 15852 waits for ShareLock on transaction 998; blocked by process 3188. Process 3188 waits for ShareLock on transaction 999; blocked by process 15852. SQL state: 40P01. Detail: Process 15852 waits for ShareLock on transaction 998; blocked by process 3188. Process 3188 waits for ShareLock on transaction 999; blocked by process 15852. Hint: See server log for query details. Context: while locking tuple (5,12) in relation "film".

Query

```
--Step 1
1 BEGIN ;
2
3 UPDATE film SET rental_rate = 2.99
4 WHERE film_id = 100;
5
6
7 --Step 3
8 UPDATE film SET rental_rate = 3.99
9 WHERE film_id = 200;
10 --Blocks ( Waiting for B)
```

Query History

Scratch Pad

Data Output

Messages

Notifications

UPDATE 1

Query returned successfully in 13 secs 201 msec.

Total rows: Query complete 00:00:13.201 CRLF Ln 8, Col 21

Query

```
--Step 2
1 BEGIN ;
2
3 UPDATE film SET rental_rate = 2.99
4 WHERE film_id = 200;
5
6
7 --Step 4
8 UPDATE film SET rental_rate = 3.99
9 WHERE film_id = 100;
10 --BOOM! ERROR : deadlock detected
```

Query History

Scratch Pad

Data Output

Messages

Notifications

ERROR: deadlock detected
Process 15852 waits for ShareLock on transaction 998; blocked by process 3188.
Process 3188 waits for ShareLock on transaction 999; blocked by process 15852.

SQL state: 40P01
Detail: Process 15852 waits for ShareLock on transaction 998; blocked by process 3188.
Process 3188 waits for ShareLock on transaction 999; blocked by process 15852.
Hint: See server log for query details.
Context: while locking tuple (5,12) in relation "film"

Total rows: Query complete 00:00:01.084 CRLF Ln 8, Col 21

Lab 2: Replicating Non-Repeatable read

The screenshot displays two side-by-side PostgreSQL query editors. The left editor shows a sequence of SQL statements: Step 1 (BEGIN), Step 2 (SELECT rental_rate FROM film WHERE film_id = 10;), Step 3 (SELECT rental_rate FROM film WHERE film_id = 10;), and Step 4 (COMMIT;). The right editor shows the same sequence, but with a COMMIT statement. The left editor also shows a table view of the rental_rate column, with a value of 4.99.

Query

```
--Step 1
1 BEGIN ;
2
3 SELECT rental_rate FROM film
4 WHERE film_id = 10;
5 --Result : 4.99 (S-Lock released)
6
7 --Step 3
8 SELECT rental_rate FROM film
9 WHERE film_id = 10;
10 --NEW RESULT : 0.99!
11 COMMIT;
12 select rental_rate
13 from film where film_id=10;
```

Query History

Scratch Pad

Data Output

Messages

Notifications

Showing rows: 1 to 1 Page No: 1 of 1

rental_rate
4.99

Total rows: 1 Query complete 00:00:00.132 CRLF Ln 8, Col 2

Query

```
--Step 2
1 BEGIN;
2
3 UPDATE film SET rental_rate = 0.99
4 WHERE film_id = 10;
5
6
7 COMMIT;
8 --Modifies data and commits
9 --while A is still active .
```

Query History

Scratch Pad

Data Output

Messages

Notifications

COMMIT

Query returned successfully in 108 msec.

Total rows: Query complete 00:00:00.108 CRLF Ln 6, Col 1

Query

Query History

Scratch Pad X

Query

Query History

Scratch Pad X

```

1  --Step 1
2  BEGIN ;
3  SELECT rental_rate FROM film
4  WHERE film_id = 10;
5  --Result : 4.99 (S-Lock released)
6
7  --Step 3
8  SELECT rental_rate FROM film
9  WHERE film_id = 10;
10 --NEW RESULT : 0.99!
11 COMMIT;
12 select rental_rate
13 from film where film_id=10;

```

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No: 1 of 1

rental_rate	numeric (4,2)
1	0.99

COMMIT

Query returned successfully in 108 msec.

✓ Successfully run. Total query runtime: 171 msec. 1 rows affected. ✕

Total rows: 1

Query complete 00:00:00.171

CRLF

Ln 9, Col 20

Total rows: 1

Query complete 00:00:00.108

CRLF

Ln 6, Col 1

Lab 4: Preventing Race Conditions

Query

Query History

Scratch Pad X

Query

Query History

Scratch Pad X

```

1  --Step 1
2  BEGIN;
3  SELECT * FROM inventory
4  WHERE inventory_id = 10 FOR UPDATE;
5  --Locks the row proactively
6
7  --Step 3
8  INSERT INTO rental
9  (rental_date,inventory_id,customer_id,staff_id)
10 VALUES (NOW() , 10 , 1 , 1 ) ;
11 COMMIT;

```

Data Output

Messages

Notifications

Showing rows: 1 to 1

Page No: 1 of 1

inventory_id	film_id	store_id	lastUpdate
[PK] integer	smallint	smallint	timestamp without time zone
1	10	2	2006-02-15 10:09:17

COMMIT

Query returned successfully in 108 msec.

✓ Successfully run. Total query runtime: 132 msec. 1 rows affected. ✕

Total rows: 1

Query complete 00:00:00.132

CRLF

Ln 3, Col 1

Total rows: 1

Query complete 00:00:00.108

CRLF

Ln 2, Col 6

QueryQuery History

```
1 --Step 1
2 BEGIN;
3 SELECT * FROM inventory
4 WHERE inventory_id = 10 FOR UPDATE;
5 --Locks the row proactively
6
7 --Step 3
8 INSERT INTO rental
9 (rental_date,inventory_id,customer_id,staff_id)
10 VALUES (NOW() , 10 , 1 , 1 ) ;
11 COMMIT;
```

Data OutputMessagesNotifications

Showing rows: 1 to 1 of 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

Total rows: 1Query complete 00:00:00.132CRLF Ln 3, Col 1

QueryQuery History

```
1 --Step 2
2 BEGIN;
3 SELECT * FROM inventory
4 WHERE inventory_id = 10 FOR UPDATE ;
5 --BLOCKS IMMEDIATELY
6
7 --Step 4
8 --Unblocks after A commits .
9 --App logic detects i t 's already rented !
10 ROLLBACK ;
11
```

Data OutputMessagesNotifications

Showing rows: 1 to 1 of 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

Total rows: Waiting for the query to complete... 00:00:05.066CRLF Ln 2, Col 1

3:04 PM9/25/2026

QueryQuery History

```
1 --Step 1
2 BEGIN;
3 SELECT * FROM inventory
4 WHERE inventory_id = 10 FOR UPDATE;
5 --Locks the row proactively
6
7 --Step 3
8 INSERT INTO rental
9 (rental_date,inventory_id,customer_id,staff_id)
10 VALUES (NOW() , 10 , 1 , 1 ) ;
11 COMMIT;
```

Data OutputMessagesNotifications

ERROR: column "status" of relation "inventory" does not exist
LINE 2: set status = 'currently rented'
^
QUERY: update inventory
set status = 'currently rented'
where inventory_id = new.inventory_id
CONTEXT: PL/pgSQL function check_inventory() line 7 at SQL statement

SQL state: 42703

Total rows: 1Query complete 00:00:00.128CRLF Ln 11, Col 8

QueryQuery History

```
1 --Step 2
2 BEGIN;
3 SELECT * FROM inventory
4 WHERE inventory_id = 10 FOR UPDATE ;
5 --BLOCKS IMMEDIATELY
6
7 --Step 4
8 --Unblocks after A commits .
9 --App logic detects i t 's already rented !
10 ROLLBACK ;
11
```

Data OutputMessagesNotifications

Showing rows: 1 to 1 of 1

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

Total rows: 1Query complete 00:00:15.915CRLF Ln 2, Col 1

3:04 PM9/25/2026

